

PyPedal — Python Software for Pedigree Analysis

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1 Preface

1.1 Acknowledgments

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The Newfoundland pedigree was taken from the [Newfoundland Dog database](#) and is used with permission.

The pedigree of European royalty used in the GEDCOM discussion, [ged3.ged](#), was taken from the [Genealogy Forum website](#). It is believed to be in the public domain, and is used with the knowledge of the website administrators.

2 Introduction

i Any sufficiently disguised bug is indistinguishable from a feature. — Rich Kulawiec

2.1 What is PyPedal?

This chapter introduces PyPedal, a Python-language tool for working with pedigrees.

PyPedal (**P**ython **P**edigree **A**nalysis) is a tool for analyzing pedigree files. It calculates several quantitative measures of genetic diversity from pedigrees, including average coefficients of inbreeding and relationship, effective founder numbers, and effective ancestor numbers. Checks are performed catch common mistakes in pedigree files, such as parents with more recent birthdates or smaller ID numbers than their offspring and animals appearing as both sires and dams in the pedigree. Tools for pedigree visualization and report generation are also provided. PyPedal only makes use of information on pedigree structure, not individual genotypes. Genotypes can be assigned to founders for use in gene-dropping simulations to calculate the effective number of founder genomes.

PyPedal is a [Python](#) language module that may be called by programs or used interactively. You must have Python 3.11 (or later) installed in order to use PyPedal. The NumPy module must also be installed in order for you to use PyPedal. In addition, there are several third-party packages used by PyPedal, which are discussed in the chapter on installing PyPedal.

This manual is the official documentation for PyPedal. It includes a tutorial and is the most authoritative source of information about PyPedal with the exception of the source code. The tutorial material will walk you through a series of manipulations of a simple pedigree. All users of PyPedal are encouraged to follow the tutorial with a working installation.

2.2 Features

Key features of PyPedal include:

- Reading pedigree files in user-defined formats;
- Checking pedigree integrity (duplicate IDs, parents younger than offspring, etc.);
- Generating summary information such as frequency of appearance in the pedigree file;

- Reordering and renumbering of pedigree files;
- Computation of the numerator relationship matrix (A) from a pedigree file using the tabular method;
- Inbreeding and relationship calculations for large pedigrees (pedigrees up to 600,000 animals have been processed with PyPedal);
- Computation of coefficients of inbreeding using the Meuwissen and Luo (1992) and modified Meuwissen and Luo (Quaas, 1995, unpublished) algorithms;
- Computation of average total and individual coefficients of inbreeding and relationship;
- Calculation of coefficients of partial inbreeding using an iterative tabular method (Lacy, Alaks, and Walsh 1996; Gulisija et al. 2006);
- Calculation of coefficients of ancestral inbreeding using the methods of Ballou (1997) or Suwanlee et al. (2007);
- Decomposition of A into T and D such that $A = TDT'$;
- Computation of the direct inverse of A (not accounting for inbreeding) using the method of Henderson (1976);
- Computation of the direct inverse of A (accounting for inbreeding) using the method of Quaas (1976);
- Storage of A and its inverse between user sessions as persistent Python objects using the `pickle` module to avoid unnecessary calculations;
- Calculation of theoretical and actual effective population sizes;
- Computation of effective founder number using the exact algorithm of Lacy (1989);
- Computation of effective founder number using the approximate algorithm of Boichard et al. (1997);
- Computation of effective ancestor number using the algorithms of Boichard et al. (1997);
- Selection of subpedigrees containing all ancestors of an animal;
- Identification of the common relatives of two animals;
- Calculation of the inbreeding of offspring from a prospective mating;
- Output to ASCII text files, including matrices, coefficients of inbreeding and relationship, and summary information;
- Importation and exportation of GEDCOM 5.5 files;
- Importation and exportation of GENES 1.20 (dBase III) files;
- Loading pedigrees from, and saving them to, MySQL, Postgres, and SQLite databases;
- Simulation of pedigrees using an algorithm derived from that used in Matvec 1.1a;
- Set operations on pedigrees (unions and intersections);

PyPedal has been used to perform calculations on pedigrees as large as 600,000 animals and has been used in scientific research (Cole, Franke, and Leighton 2004; Cole 2007).

2.3 Where to get information and code

PyPedal and its documentation are available from the [PyPedal 3 Github repository](#).

3 Getting Started

For an example of how to load a pedigree, see Listing 3.1.

Listing 3.1

```
from PyPedal import pyp_newclasses
options = {
    'renumber': 0,
    'pedfile': '../test_files/mrode.ped',
    'pedformat': 'asd',
    'pedname': 'Mrode Pedigree Table 2.1',
    'pedigree_is_renumbered': 1,
    'messages': 'quiet',
}
result = pyp_newclasses.load_pedigree(options)
result.metadata.printme()
```

Metadata for Mrode Pedigree Table 2.1 (../test_files/mrode.ped)

Records:	6
Unique Sires:	3
Unique Dams:	2
Unique Gens:	1
Unique Years:	1
Unique Founders:	2
Unique Herds:	1
Pedigree Code:	asd

References

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